HYUNWOO KIM

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RESEARCH INTEREST

ML for Quantum and Quantum for ML | Quantum Machine Learning | Quantum Generative Models | Quantum Error Mitigation | Pauli Propagation and Classical Surrogates | Trainability and Expressivity of VQCs | Industrial NISQ Algorithms

EDUCATION

BACHELOR OF SCIENCE: PHYSICS & ARTIFICIAL INTELLIGENCE
Sogang University 03/2020 to 02/2026; *Seoul*

- **GPA**: 3.81/4.30 (95.8 percentile; cohort average: 91.4)
- **Honors: Magna Cum Laude**
- **Relevant Physics Coursework**: Quantum Physics, Modern Optics, Statistical Physics, Electromagnetics, Solid-State Physics, Experimental Physics, Classical Dynamics, Thermodynamics, Computational Physics II
    - External coursework: Quantum Information Science, Quantum Computing at Yonsei University via credit exchange
- **Relevant AI & CS Coursework**: Data Structures, Algorithms, Statistics, Linear Algebra, Introduction to Machine Learning, Introduction to Deep Learning, Computational Thinking (Python), Introduction to C Language

EXPERIENCE

NORMA, Inc.
Researcher, Quantum AI Team 07/2025 to Present; *Seoul*

- **Implemented quantum generative models** (QGAN, IQP-QCBM) using Qiskit, PennyLane and NVIDIA CUDA-Q to study expressive power and trainability of variational quantum circuits in generative modeling
- **Developed PadoPauli**, a **Python toolkit for Pauli Propagation**, enabling efficient simulation of operator evolution and implementation of diverse quantum algorithms, including QAOA, VQE, and QEM
- **Analyzed noise resilience of variational quantum circuits (VQCs),** including VQE and QAOA, under simulation and real-hardware environments
- **Developed a neural-network-based QEM (Quantum Error Mitigation) model** to suppress noise effects in small scale variational quantum circuits

DEPARTMENT OF PHYSICS, POHANG UNIVERSITY OF SCIENCE AND TECHNOLOGY
Research Assistant (Supervisor: Prof. Yoonho Kim) 07/2024 to 07/2024; *Pohang*

- Research Intern at Quantum Optics and Quantum Information Group (Prof. Yoonho Kim)
- **Designed and aligned a high-precision interferometer and conducted Bell test measurements** using a 1550 nm Kalmár laser for quantum optics research
- **Performed SPDC (Spontaneous Parametric Down-Conversion) experiments** for photon-pair generation and entanglement studies
- **Supported time-energy entanglement experiments**, contributing to data collection and experimental setup optimization

DEPARTMENT OF PHYSICS, SOGANG UNIVERSITY
Research Assistant (Supervisor: Prof. Minsoo Kim) 09/2024 to 12/2024; *Seoul*

- **Analyzed quantum Hall transport data** in mesoscopic 2D systems
- **Studied valley-dependent transport and Dirac-cone** related phenomena
- **Supported theoretical modeling** linking experimental observables to band structure

PUBLICATION & PRESENTATION

Preprint / Accepted Paper

1) **[H. Kim, Y. Lee], "SAFE ma-QAOA: Surrogate-Assisted and Fine-Tuning Enhanced Multi-Angle QAOA with Parameter Distillation" (2026).** arXiv: 2605.23377.
   - Accepted to the Technical Papers Track (Quantum Algorithms, QALG), IEEE International Conference on Quantum Computing and Engineering (QCE) 2026 — Sep 13-18, 2026; *Toronto*
   - Invited Talk, Waterloo Quantum Club (WQC), University of Waterloo — Jul 15, 2026; *Online*
   - Poster, Quantum Future Korea 2026 (2nd National Undergraduate Quantum Conference) — May 15, 2026; *Seoul*

Posters & Presentations

2) **[H. Kim, Y. Lee], "PadoPauli: A GPU-accelerated Implementation of Pauli Propagation Surrogate" (2026).**
   - Poster, Conference on Quantum Information (Quantum Korea 2026) — Jul 01-03, 2026; *Seoul*
   - Code: github.com/Norma-Q/Pauli-Propagation---GPU-acceleration

3) **[H. Kim, Y. Lee], "Two Approaches of Potential Quantum Advantage with Quantum Generative Models in Drug Discovery" (2026).**
   - Poster, 1st National Undergraduate Quantum Conference — Feb 28, 2026; *Seoul*
   - Poster, Annual Meeting of the Quantum Information Society of Korea — Feb 25-27, 2026; *Seoul*

SELECTED PROJECTS

PadoPauli: A GPU-accelerated SDK for Pauli Propagation Surrogate
- Developed a GPU-based, differentiable Pauli Propagation SDK with support for both CUDA and ROCm backends
- Achieved state-of-the-art performance in benchmarks against pauli-prop (qiskit), PauliPropagation (Julia), and cuPauliProp (NVIDIA)
- Currently extending to NPU-based simulation for further acceleration

Realizing Quantum Advantage in the Generation of Drug Library by Quantum Machine Learning
- Conducted research on quantum generative models for molecular drug library generation, integrating **KRAS G12D-binding ligand candidates** into the evaluation pipeline
- A 2024-2026 National Research Foundation of Korea Quantum Computing Challenge Project

Full-stack Remote Control System for Laboratory Instrumentation
- Developed a **Python-based remote-control system** for digital oscilloscopes to automate experimental data acquisition
- Implemented end-to-end functionality including **device control, data extraction, real-time visualization and analysis** using Python

HONORS & AWARDS

Full Tuition Scholarship (Sogang University, 2020-2025)
- Awarded a merit-based scholarship covering four years of full tuition, granted based on academic performance and GPA criteria

Merit-Based Scholarship
- **Sogang Alumni Association Scholarship** (2024): Stipend-based scholarship awarded through academic excellence and personal statement evaluation
- **Catholic Outstanding Student Scholarship** (2025): Stipend-based scholarship awarded based on academic achievement and personal evaluation

Residential Scholarship
- **Accommodation Scholarship** (Lotte Scholarship Foundation, 08/2023 – 07/2025): Two-year fully funded accommodation awarded through academic and personal evaluation

TEACHING & OUTREACH

CONFERENCE OUTREACH (NORMA, Inc.)

- Q2B 2026 — operated NORMA's technical demo booth; presented PadoPauli use case to industry researchers and visitors Jun 04-05, 2026; *Tokyo*
- Quantum Korea 2026 — booth operation & technical presentation of quantum computing solutions to attendees Jul 01-03, 2026; *Seoul*

1ST QUANTUM AI COMPETITION (NORMA, Inc. & AI Factory)

Technical Mentor Aug 18-19, 2025; *Jeonju*

- Mentored participating teams on quantum algorithm design and quantum machine learning implementation (55 teams entered; 10 finalists)

INSTITUTE FOR CONVERGENCE EDUCATION, SOGANG UNIVERSITY

Teaching Assistant 03/2024 to 06/2025; *Seoul*

- Lecture: Introduction to Artificial Intelligence Programming (Python)
- Instructed undergraduate students on core Python programming principles and problem-solving techniques
- Coordinated with the professor and other TAs to prepare course materials, effectively addressing student inquiries, debugging code, and assisting with practical exercises

TRAINING

Seoul National University 07/2024 to 08/2025

- The Basics of Quantum Computing and Information

LG AI Institute 01/2024 to 02/2024

- Acquired expertise in advanced machine learning techniques
- Developed AI model that predicts successful sales conversion using Marketing Qualified Lead (MQL) customer information

TECHNICAL SKILLS

- **Experimental Physics:** Interferometry, SPDC, Hall effect
- **Quantum SDKs**: Qiskit, PennyLane, NVIDIA CUDA-Q, Quantinuum Nexus
- **Programming**: Python (NumPy, SciPy, PyTorch), C

ADDITIONAL EXPERIENCE

MILITARY POLICE, REPUBLIC OF KOREA, ARMY

SDT (Special Duty Team) Sniper 02/2021 to 08/2022; *Gangwon-do*

- Served as a sniper for SDT (Special Duty Team) responsible for counterterrorism and commander's security
- Managed and calibrated precision optical equipment and weaponry for optimal operational readiness